

PROFILE SUMMARY

AI Engineer / Data Engineer with 2.6+ years of professional experience and 5+ years of coding expertise, specializing in building production-grade ML systems and data platforms. Experienced in designing end-to-end pipelines from data ingestion to model deployment using Spark, Kafka, and modern cloud-native architectures on AWS. Hands-on with MLOps and LLM systems, including model deployment, scalable APIs, and multi-agent workflows, with strong foundations in data engineering (Airflow, dbt, Snowflake, BigQuery, PostgreSQL). Proven ability to develop reliable, high-performance systems on Kubernetes and serverless environments with a focus on scalability, observability, and real-world impact.

SKILLS

Machine Learning & AI: Supervised/Unsupervised Learning, Deep Learning, NLP, Generative AI, LLMs (GPT, LLaMA, Gemini, Falcon), Time Series Analysis, Computer Vision (OpenCV), Prompt Engineering, Fine-tuning (LoRA/QLoRA)

LLM Engineering & Frameworks: LangChain, LlamaIndex, RAG (Retrieval-Augmented Generation), LangGraph, AutoGen, LangSmith, Vector Databases (Pinecone, FAISS, ChromaDB), OpenAI API, Hugging Face Transformers

MLOps & LLMops: MLflow, Model Deployment, CI/CD Pipelines, Experiment Tracking, Model Monitoring, Orchestration (Airflow), AIOps, Scalable ML Systems

Data Engineering & Big Data: Apache Spark (PySpark, SQL, Structured Streaming), Apache Kafka, AWS Glue, Delta Lake, Parquet, dbt

Cloud Platforms: AWS (SageMaker, Bedrock, S3, Lambda, API Gateway, DynamoDB, Cognito, SNS, CloudWatch, IAM), Azure (Azure ML, App Services), GCP (Vertex AI, Cloud Functions)

Backend & APIs: Python (FastAPI, Flask), Java (Spring Boot), REST APIs, GraphQL, Microservices Architecture

Databases & Warehousing: Snowflake, Google BigQuery, PostgreSQL/PostGIS, MySQL, SQL Server, MongoDB, DynamoDB

Programming: Python, Java, Kotlin, C++, JavaScript/TypeScript, SQL, R

Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, Plotly

DevOps & Tools: Docker, Kubernetes, GitHub Actions, GitLab CI/CD, Terraform (basics), Gradle, Postman, Amplify

PROJECTS

MultiAgent Route & Weather System — GeoAI Multi-Agent Orchestration GitHub

Python, AutoGen, FastAPI, Google Maps APIs, PostGIS (Docker), Open-Meteo, Ollama

- Designed a **multi-agent LLM system** orchestrating route planning, geospatial risk analysis, and weather intelligence using tool-calling workflows.
- Integrated **Google Maps APIs** to generate encoded polylines and route legs; enabled downstream spatial analytics.
- Implemented **PostGIS-based accident analysis** using buffered polyline queries via custom SQL functions.
- Built **weather enrichment pipeline** sampling real-time conditions along route coordinates using Open-Meteo API.

BirdTag — Serverless Media Tagging & Search Platform GitHub | Live

AWS (S3, Lambda, API Gateway, DynamoDB, Cognito, SNS), Python, REST

- Designed **event-driven ingestion pipeline** using S3 triggers and Lambda for thumbnail generation, metadata extraction, and tagging.
- Built scalable **REST APIs** for search and retrieval; integrated Cognito authentication and SNS notifications.
- Improved reliability via **CloudWatch monitoring**, optimized Lambda memory/timeout, and reduced cold-start latency.

Big Data Pipelines — Streaming & Batch Analytics GitHub

Apache Kafka, Spark Structured Streaming, Python, Parquet/Delta, Airflow, dbt

- Built Kafka producers and Spark streaming consumers performing **windowed aggregations** (6h/daily) with watermarks.
- Optimized storage using **partitioned Parquet/Delta formats**, checkpoints, and idempotent writes.
- Designed **Airflow DAGs and dbt models** for scalable ELT workflows with monitoring and backfill strategies.

EXPERIENCE

Infor

04/2022 – 07/2024

Associate Software Engineer — EPM Team

- Automated deployments for reporting apps on Infor CloudSuite using **GitLab CI/CD**, **Docker**, and YAML pipelines across Linux environments.
- Developed Python utilities to generate/archive report snapshots in **AWS S3**, improving auditability and recovery.
- Managed infra with **EC2 ASG**, Launch Templates, ECR, Secrets Manager; implemented observability via **CloudWatch** and **SNS**.
- Integrated with Infor Document Management (IDM); enforced least-privilege IAM and role-based access.
- Participated in Agile sprints; coordinated releases with platform/ops to reduce lead time and deployment risk.

EDUCATION

Monash University, Melbourne

06/2024 – Present

Master of Information Technology

Jawaharlal Nehru Technological University

08/2018 – 06/2022

B.Tech. in Computer Science and Engineering